CLAIM_ALPHA_CLOSURE_GATE_v39

Title: P0mu_R as RT's alpha-facing closure-gate coefficient

Author: Bo Fremling

Project: Rhythm Theory (RT)

Status: current public alpha-facing claim; supersedes v38.4/v38.5 wording

Date: 2026-06-10

Scope: internal RT coefficient first; physical alpha only as conditional audit

Canonical public PDF after publication:

https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/CLAIM_ALPHA_CLOSURE_GATE_v39.pdf

Repository / Release branch:

https://github.com/Bo-Fremling/rhythm-theory-release/tree/Release

Release archive page:

https://github.com/Bo-Fremling/rhythm-theory-release/releases

Public geometry companion:

https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_GEOMETRIC_CARRIER_MODEL_FOR_P0MU_R_v1.pdf

> *This PDF is intended to stand alone as the current public alpha-facing claim. The geometry companion is public background for the carrier/readout geometry, not a private handoff requirement.*

0. One-line current claim

```
P0mu_R[Theta] is the frozen internal RT same-port scalar coefficient.
alpha(0) is the conditional external audit, not the theorem object.
```

This file preserves the `CLAIM_ALPHA_CLOSURE_GATE_v*` public naming path so older texts that point readers to the latest `CLAIM_ALPHA_CLOSURE_GATE_v*` land on the current state.

`v39` supersedes `v38.4`/`v38.5` as current wording. The numerical kernel is not discarded; the public interpretation is narrowed from a direct alpha-claim to an internal RT coefficient plus conditional audit.

1. What is claimed

The theorem object is:

```
P0mu_R[Theta]
```

It is an internal RT static scalar same-port coefficient built from the RT-RP readout ledger.

After the separate Single-Z2 character bridge:

```
Theta_AB = pi
```

the coefficient has the numerical audit value:

```
lambda_P0mu_R(pi) = pi/420 * 3020233/3095820
                  ~= 0.007297352563048502
1/lambda_P0mu_R(pi) ~= 137.03599920109184
```

This value is an audit of the frozen ledger. It is not an input to the construction.

2. What is not claimed

This document does not claim a full derivation of:

```
electromagnetism
QED
the Coulomb law
the Standard Model
the running of alpha
masses
mixing angles
gauge dynamics
gravity
transition amplitudes
```

The physical alpha comparison is valid only under the Same-Port EM Interface Contract:

```
physical low-energy EM must correspond to RT's low-energy, static,
scalar, pair-wise, boundary-readable, unit-source-normalized,
object-independent, kernel-stripped same-port interface.
```

If that interface fails, `P0mu_R` remains an internal RT coefficient and is not physical alpha.

3. RT-RP rules used by the ledger

The following RT-RP rules carry the structural load. They are not optional decoration.

```
R0. RP is internal readout, not external subtraction.
R1. A raw PP difference is latent until it becomes an activated readout relation.
R2. The first visible scalar port must have a balance-null state.
R3. The first deviation from null must pass through a latent sigma-guard.
R4. The first visible primitive scalar surface witness must have near >= 3.
R5. Local-first/no-skip selects the first valid local port.
R6. Current and retained trace must be joined through an activated seam.
R7. Numerical comparison is performed only after the ledger is frozen.
```

These rules force the following internal chain if RT-RP is accepted:

```
null -> sigma-guard -> first visible primitive port
5|5  -> 4|6         -> 3|7
rho = 10
```

If direct external readout is allowed, then competing branches such as `rho=8 -> 3|5` return. That is the main internal ontology failure point.

4. Register chain

From the first visible port:

```
rho = 10
split class = {3,7}
```

The branch return is:

```
R = lcm(3,7) = 21
```

The surface grid is:

```
C30 = Z10_port x Z3_surface = 30
```

The first common boundary-address register is:

```
B = lcm(21,30) = 210
```

The common anchor overlap is:

```
Q = gcd(21,30) = 3
```

The current/retained surface support after anchor pushout is:

```
G = 2R - Q = 42 - 3 = 39
```

The internal RT-RP carrier capacity is:

```
C = C30 x Z3_trace = 30 x 3 = 90
```

Meaning:

```
Z3_trace = current / retained / activated seam
```

`C60 = current + retained` is an external-subtraction alternative. Under RT-RP, the activated seam is required, so `C90` is used.

5. Geometric carrier status

The words `surface`, `seam`, `port`, `trace`, `current`, `retained`, and `activated seam` refer to a typed discrete carrier/readout geometry:

```
C = Z10_port x Z3_surface x Z3_trace
  = 10 x 3 x 3
  = 90 carrier cells
```

This can be represented as a discrete 10-by-3-by-3 carrier block with coordinates:

```
(u, s, t)
u in Z10_port      local port coordinate
s in Z3_surface   surface stencil
t in Z3_trace     current / retained / activated seam
```

This is a real discrete carrier model, but it is not yet a full smooth metric 3D physics model. The public companion file gives the geometric carrier picture behind the ledger:

```
RT_GEOMETRIC_CARRIER_MODEL_FOR_P0MU_R_v1.pdf
https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_GEOMETRIC_CARRIER_MODEL_FOR_P0MU_R_v1.pdf
```

6. Coefficient ledger

The P0 base load is:

```
Delta_0 = 2 + 2/rho - 1/(rho*G)
```

Term roles:

```
2         two relational endpoint ranks
2/rho     local port exposure at both endpoints
1/(rhoG) rank-one scalar seam credit
```

The `mu_R` completion makes the latent seam-core boundary-readable without changing the world-registers:

```
1/(rhoG) -> (1 + epsilon)/(rhoG)
```

with:

```
active_R = (R - 1)/R = 20/21
epsilon  = active_R/B = (20/21)/210 = 2/441
```

Thus:

```
Delta_mu = 2 + 2/rho - (1 + epsilon)/(rho*G)
```

and:

```
U_mu = 1 - Delta_mu/C
```

The frozen coefficient is:

```
lambda_P0mu_R[Theta] = Theta_AB/(2B) * U_mu
```

With:

```
rho = 10
R   = 21
B   = 210
G   = 39
C   = 90
epsilon = 2/441
```

one obtains:

```
Delta_mu = 75587/34398
U_mu     = 3020233/3095820
lambda_P0mu_R[Theta] = Theta_AB/420 * 3020233/3095820
```

After the separate character bridge `Theta_AB = pi`:

```
lambda_P0mu_R(pi) = pi/420 * 3020233/3095820
```

7. Relation to v38.4

The v38.4 numerical expression and v39 expression are algebraically equivalent at the kernel level. The change is the public typing.

```
v38.4: alpha-facing closure gate, too easy to read as direct physical alpha.
v39:   P0mu_R[Theta] is the internal theorem object;
       alpha(0) is the conditional audit.
```

This is why v39 is a semantic update rather than a small formatting patch.

8. Failure points

The main failure points are explicit:

```
F1. RT-RP internal readout is wrong or insufficient.
F2. Sigma-guard is not forced.
F3. rho=8 direct readout is allowed.
F4. C90 is not the right carrier domain; C60 or another domain wins.
F5. G39 is not the right current/retained support valuation.
F6. mu_R / active_R = (R-1)/R is not the right completion.
F7. Theta_AB = pi is not the correct Single-Z2 character bridge.
F8. q_RT is not the physical EM source grammar.
F9. Physical EM is not RT's same-port scalar boundary interface.
F10. A hidden scale factor kappa is needed.
F11. The low-energy alpha(0) target is the wrong physical comparison.
```

Any one of these can break the physical alpha identification.

9. Final status

```
Strong internal result:
  A frozen RT-RP ledger coefficient P0mu_R[Theta].

Strong numerical audit:
  P0mu_R(pi) is extremely close to alpha(0).

Not yet achieved:
  A full derivation of physical electromagnetism or the Standard Model.

Main next target:
  Show that RT-RP same-port boundary readout is physically the low-energy EM scalar pair interface.
```

Short form:

```
P0mu_R is the theorem object.
alpha is the audit.
The geometry companion explains the carrier.
The EM bridge remains the decisive open point.
```